Eli Whitaker

Oakland, CA | (916) 284-3246 | eli.p.p.whitaker@gmail.com | eliwhitaker.org | github.com/rekatihwile

Summary

Mechanical Engineering graduate from UCLA (4.000 GPA, Summa Cum Laude) focused on robotic system integration: sensors, embedded motion hardware, Linux-based compute, controls, and physical testing. Built autonomous prototypes that combine cameras, microcontrollers, Jetson/Linux compute, serial/CAN-style communication, ROS 2, Python/C++, calibration, visual servoing, and failure recovery on real hardware. Strong fit for hands-on R&D integration work where prototypes must be debugged as complete systems rather than isolated parts.

Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. Mechanical Engineering | GPA: 4.000 | Summa Cum Laude — *June 2026*

- Relevant Coursework: Robotics, Control of Robotic Systems, Dynamic Robotic Systems, Dynamic Systems and Control, Mechatronics/Capstone Design, Electrical and Electronic Circuits, Circuit Measurements Lab, Manufacturing Processes.

Technical Skills

- **Robotics / Autonomy:** system integration, sensor calibration, camera-to-robot transforms, stereo vision, path planning, state machines, visual servoing, inverse kinematics, PD/PID control, hardware bring-up.
- **Embedded / Interfaces:** Teensy 4.1, Jetson AGX Orin, stepper drivers, servo PWM, Dynamixel actuators, GRBL gantry control, UART/serial, CAN-style rover control, USB cameras, Ethernet/networked debugging.
- **Software / Tools:** Python, C++, MATLAB, ROS 2, Linux CLI, Git/GitHub, OpenCV, NumPy, SciPy, PyTorch/YOLO workflows, browser-based local dashboards, LaTeX documentation.
- **Hardware / Test:** prototype assembly, wiring, sensor mounts, 80/20 frames, 3D-printed fixtures, actuator tuning, calibration procedures, bench testing, trial logging, root-cause debugging, test-plan documentation.

Research Experience

Structures–Computer Interaction Lab – Robotic Laser Weeding — Aug. 2025 – Jun. 2026
Undergraduate Researcher — *UCLA*

- Integrated a full autonomous laser-weeding prototype using an AgileX SCOUT 2.0 rover, planar gantry, 60 W blue diode laser, Jetson compute, stereo USB cameras, battery power, and field-testable mechanical/electrical packaging.
- Built the perception-to-actuation stack: stereo survey capture, burst-stable target filtering, left/right matching, triangulation into gantry coordinates, short-route traversal, local target re-identification, PD visual servoing, and laser fire sequencing.
- Developed operator/debug tooling hosted on the Jetson for survey capture, class confidence, crop/window tuning, target lists, IoU thresholds, robot motion commands, and real-time integration debugging over local network access.
- Supported autonomous validation trials; decomposed runtime into travel, re-identification, fine alignment, and laser dwell to identify system-level bottlenecks before hardware changes.
- Co-authored/submitted IEEE RAL-style research package on diode-laser ablation, energy efficiency, label-efficient perception, autonomous targeting, and battery-powered field deployment constraints.

Engineering Projects

Autonomous Grocery Packing Robot — *UCLA MAE 162D/E Capstone*

- Led autonomy and control integration for a SCARA-like RRPR grocery-bagging robot with stereo camera, overhead camera, Teensy 4.1 motion controller, stepper axes, servo gripper, and Python-to-firmware command pipeline.
- Connected YOLO segmentation, RAFT-Stereo disparity, masked point-cloud reconstruction, height-indexed homography, camera-to-robot calibration, inverse kinematics, and deterministic 2.5D packing into one closed-loop system.
- Implemented missed-pick recovery using post-pick re-survey, class/position/IoU/area checks, release cancellation, Z-axis re-home, and retry logic to prevent false bag-state updates after grasp failures.
- Demonstrated autonomous survey, pick, place, bag-state update, and recovery behavior on physical hardware; achieved 95.1% pick-and-place success across 425 physical attempts.

Vision-Guided Ball-Catching Robot — *UCLA MAE C163C Course Project*

- Contributed to a ROS 2-based planar 2R Dynamixel robot using a high-speed stereo global-shutter camera, OpenCV ball detection, stereo depth, IIR filtering, and real-time landing-point prediction.
- Designed and evaluated the operational-space inverse dynamics controller; analyzed singularity sensitivity, actuator limits, task-space gain tuning, step responses, and hardware instability during real robot tests.

Additional Experience

UCLA Recreation / John Wooden Center — Summer 2025 – Jun. 2026
FITWELL Supervisor (promoted from Consultant) — *Los Angeles, CA*

- Coordinated daily operations, issue resolution, maintenance tickets, and staff/patron communication in a high-traffic university facility; trusted with supervisor responsibilities during senior year.